ScreenO

documentation

1. Database Connection and Queries

Retrieves application usage limits and usage durations from a database.

Logs active application usage into the database.

Logs and retrieves usernames and passwords to the database.

2. Application Termination

Terminates applications that exceed their usage limit by killing their processes.

3. Active Application Tracking

Monitors the currently active application.

Records focus start time and calculates duration when switching apps.

4. UI Updates

Dynamically updates the FlowLayoutPanel (flpAppCards) with application usage details.

Displays usage as a percentage with a visual progress gauge.

Updates the usage graph using a bar chart for detailed application usage over time.

5. Timers

Timer1: Logs active app usage, updates cards, and checks app limits every 25 seconds.

Timer2: Periodically (every 10 seconds) checks app usage limits and terminates if necessary.

7. Windows API Integration

Uses Windows API to retrieve the name of the active application based on its process ID.

https://www.youtube.com/watch?v=dzUiBHyaNRU&pp=ygUmaG93IHRvIHVzZSB3aW5kb3dzIGFwaSBpbiB2aXN1YWwgYmFzaWM%3D

8. Settings and Navigation

Handles navigation between the dashboard, login, and settings forms.

Passes user data (ID and username) to the settings form.

9. Data Visualization

Utilizes Guna UI components to create app cards and progress bars.

Displays app usage data with dynamic visual charts.

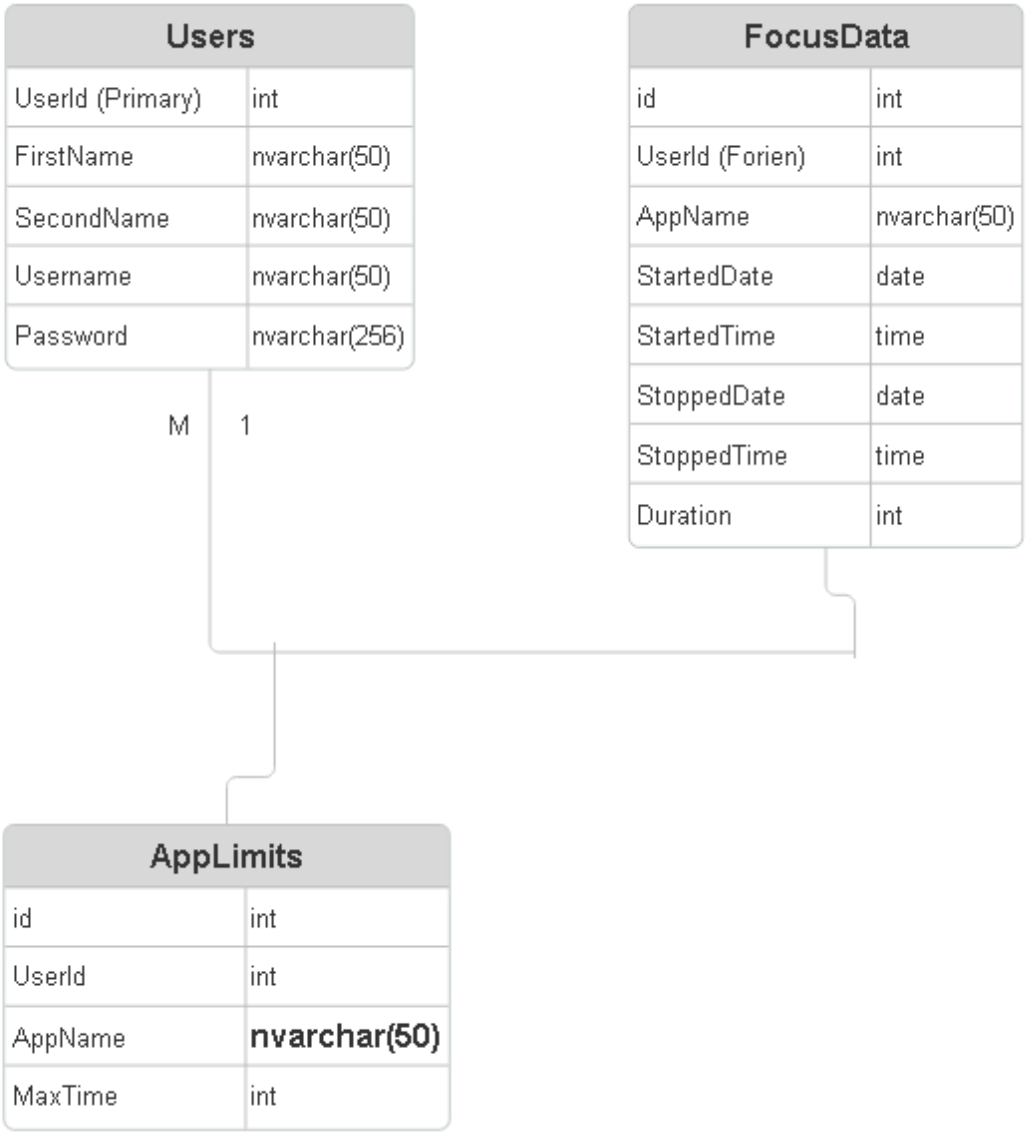
10. Custom Properties

Manages logged-in user details via LoggedInUserID and LoggedInUsername properties.

11. Utility Functions

Fetches application durations and detailed usage data from the database.

Inserts app usage logs into the database with relevant timestamps and durations.


Users
UserId (Primary)
int
FirstName
nvarchar(50)
SecondName
nvarchar(50)
Username
nvarchar(50)
Password
nvarchar(256)
M
1
FocusData
id
int
UserId (Forien)
int
AppName
nvarchar(50)
StartedDate
date
StartedTime
time
StoppedDate
date
StoppedTime
time
Duration
int
AppLimits
id
int
UserId
int
AppName
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MaxTime
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